

YEBOWEN HU

[GitHub](#) ◇ yebowen.hu@ucf.edu ◇ [Google Scholar](#)

RESEARCH INTEREST

CS PhD candidate at the University of Central Florida. My research focuses primarily on simulating large language model to imitate the human cognitive process (reasoning) and behavior (tool use, IF). Exploring methods to evaluate and enhance the LLM's capabilities in handling long and heterogeneous context under real-world scenarios.

EDUCATION

Doctor of Philosophy, University of Central Florida 2021 - Present
Major in Computer Science, GPA: 3.94 Orlando, FL
Advisors: [Fei Liu](#) and [Hassan Foroosh](#)

Master of Science, University of Florida 2016 - 2018
Major in Electrical and Computer Engineering Gainesville, FL
(minor degree: Computer Science)

Bachelor of Science, Central South University 2012 - 2016
Major in Electrical and Electronic Automation Engineering ChangSha, China

PROFESSIONAL EXPERIENCES

Graduate Research Assistant Aug 2021 - Present
University of Central Florida, CECS department Orlando, FL

- **Reasoning:** Conducted in-depth research on leveraging past experiences to guide LLM decision-making under uncertain environments and enhancing reasoning capability with heterogeneous data in real-world domains
- **Evaluation:** Explored the boundaries of LLM capabilities spanning complex instruction following, formal logic, analytical reasoning, and long-context stress tests
- **Industry experience** in agent development, multimodal SLM training, and advanced vertical AI product design. Familiar with post-training, reinforcement learning, prompt-engineering and prompt-tuning to advance the reasoning capability of LLM

GenAI Research Intern May 2025 - Oct 2025
Zoom GenAI Bellevue, WA

- Optimized AI coaching product and improved the performance of transcript analysis by 5.6% on LLM
- Implemented automatic rubric generation pipeline and AI agent framework for educational qualification analysis
- Developed MMLM(Audio+Text) evaluation framework for revenue accelerator project and enhanced MMLM's function calling performance by 6%

NLP Research Intern May 2024 - Aug 2024
Tencent America, LLC Bellevue, WA

- Led research on applying analytical and analogical reasoning in LLMs to complex domains like sports analysis and financial forecasting, prioritizing research with direct product application
- Proposed reasoning framework, DeFine, that improved 5-way market recommendation accuracy by 7% over Chain-of-Thought and DeLLMa, providing controllable and explainable justifications
- Proposed SportsGen, a novel framework that synthesizes sports narratives by modeling game dynamics and a metric, the Discounted Cumulative Accuracy, for numerical performance evaluation with tolerance

NLP Research Intern May 2023 - Aug 2023
Tencent America, LLC Bellevue, WA

- Led project on probing human preference influential factors on k -wise text comparisons with GPT-4
- Leveraged the Bradley-Terry Luce model to explore dominant preference factors characterizing LLM outputs
- Proposed InFoBench to evaluate planning and instruction following abilities of LLMs through decomposed tasks

PUBLICATIONS

When Reasoning Meets Information Aggregation: A Case Study with Sports Narratives

Yebowen Hu, Kaiqiang Song, Sangwoo Cho, Xiaoyang Wang, Wenlin Yao, Hassan Foroosh, Dong Yu, Fei Liu
Main conference of Empirical Methods in Natural Language Processing (EMNLP'24), Miami, Dec 2024.

SportsMetrics: Blending Text and Numerical Data to Understand Information Fusion in LLMs

Yebowen Hu, Kaiqiang Song, Sangwoo Cho, Xiaoyang Wang, Hassan Foroosh, Dong Yu, Fei Liu
Main conference of Association for Computational Linguistics (ACL'24), Bangkok, Thailand, Aug 2024.

InFoBench: Evaluating instruction following ability in large language models

Yiwei Qin, Kaiqiang Song, **Yebowen Hu**, Wenlin Yao, Sangwoo Cho, Xiaoyang Wang, Xuansheng Wu, Fei Liu, Pengfei Liu, Dong Yu
Findings of Association for Computational Linguistics (ACL'24), Bangkok, Thailand, Aug 2024.

DecipherPref: Analyzing Influential Factors in Human Preference Judgments via GPT-4

Yebowen Hu, Kaiqiang Song, Sangwoo Cho, Xiaoyang Wang, Hassan Foroosh, Fei Liu
Main conference of Empirical Methods in Natural Language Processing (EMNLP'23), Singapore, Dec 2023.

MeetingBank: A New Benchmark Dataset for Meeting Summarization

Yebowen Hu, Tim Ganter, Hanieh Deilamsalehy, Franck Dernoncourt, Hassan Foroosh, Fei Liu
Main conference of Association for Computational Linguistics (ACL'23), Toronto, Canada, Aug 2023.

DeFine: Enhancing LLM Decision-Making with Factor Profiles and Analogical Reasoning

Yebowen Hu, Xiaoyang Wang, Wenlin Yao, Yiming Lu, Daoan Zhang, Hassan Foroosh, Dong Yu, Fei Liu
Findings of Association for Computational Linguistics (ACL'25), Vienna, Austria, July 2025.

LiveMCP-101: Stress Testing and Diagnosing MCP-enabled Agents on Challenging Queries

Collaborative Research, preprint on arXiv, Aug 2025.

Complex logical instruction generation

Collaborative Research, preprint on arXiv, Aug 2025.

TCIA: A Task-Centric Instruction Augmentation Method for Instruction Finetuning

Collaborative Research, preprint on arXiv, Aug 2025.

SERVICES

- Serve as reviewers in top-tier academic conferences: ACL'25, ICLR'25, EMNLP'24, ACL'24, NAACL'24, JAIR'24, ACL'23, EMNLP'23, CoNLL'23, JAIR'22, TACL'22, TACL'21
- Serve as a program committee of the IEEE International Conference on Healthcare Informatics (IEEE ICHI 2024)

AWARDS

- ORCGS Doctoral Fellowship, University of Central Florida, 2021
- ECE Graduate Academic Achievement Awards Scholarship, University of Florida, 2016

MISCELLANEOUS

Technical Skills: Python, PyTorch, Natural Language Processing, Deep Learning Algorithm, Prompt Engineering

Language: English (Fluent), Mandarin (Native)

Personality: AI Enthusiast, Team-oriented Researcher, Articulate Communicator